

Object-Oriented Programming (OOP) Study Notes

1. **Class:** Blueprint for creating objects.
 2. **Object:** Instance of a class.
 3. **Encapsulation:** Wrapping data and methods together.
 4. **Abstraction:** Hiding implementation details.
 5. **Inheritance:** One class acquires properties of another.
 6. **Polymorphism:** Same interface, different behavior (overloading/overriding).
 7. **Constructor:** Initializes objects automatically.
 8. **Destructor:** Releases resources when object is destroyed.
- Advantages:** Reusability, Security, Maintainability, Modularity.